

Nucleic Acids

Lecturer 1

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Chapter I

Introduction

Chapter one: Introduction

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- 2- History of nucleic acids
- 3- Cell Types
- 4- Occurrences and nomenclature of nucleic acids

1- Why should I learn Na?

Why is it important?

- Nucleic Acids are the focus of attention because of its role in carrying and expressing genetic information.
- The *Human Genome Project* where over 90% (99.9% accuracy) of the 3.2 billion nucleotides (or letters) making up the genome in the human body has been sequenced.
- This has been the most expensive project in biology \$250 million, 13 years with some 1,100 biologist, computer scientist and technicians at 16 universities labs in 6 countries.
- The information, it is hoped will revolutionize the detection, prevention and treatment of conditions from cancer to depression to old age itself.

Genomic achievements since the Human Genome Project



2- History of nucleic acids

A SHORT HISTORY OF DNA



8000 BC

Long before the discovery of DNA, early farmers were using selective breeding to improve their crops and livestock. They kept back the best seed and offspring from their farms to begin the next generation.

1859

Charles Darwin publishes his theory of evolution through natural selection. It was only long after his death that his ideas finally became widely accepted.



1953

James Watson and Francis Crick accurately describe the molecular structure of DNA as a double helix.

1863

Gregor Mendel, a monk in Austria, first documents hereditary traits in garden peas.



1966

The genetic code is revealed. It is established that a sequence of three nucleotide bases corresponds to each of 20 amino acids in the production of proteins. Since then a further two amino acids have been discovered.



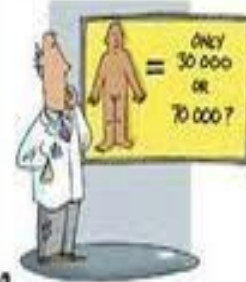
1972

The DNA composition of humans is found to be 99% similar to chimpanzees and gorillas.



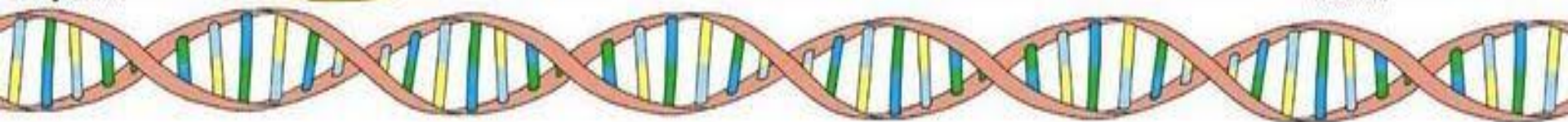
1990

The Human Genome Project is launched - an international collaborative effort to sequence the entire human genome.



2002

The Human Genome Project is completed - revealing the location of around 30 000 human genes. This number, however, is currently being debated by scientists who suggest the number is closer to 70 000.



History of nucleic acids

- In 1869 Friedrich Miescher Discovered Nucleic acids from pus cells and called it nuclein
- 1889 Richard Altmann Nuclein was shown to have acidic properties, hence it became called **nucleic acid**



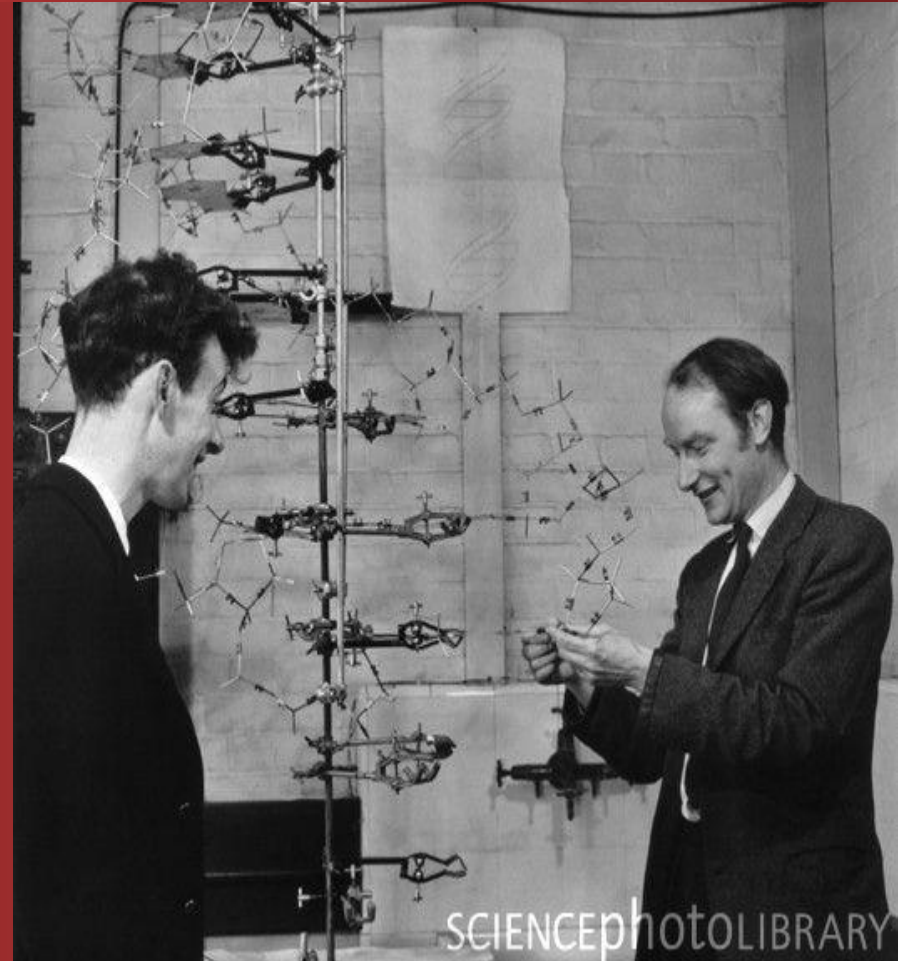
History of nucleic acids

- In 1938 Astbury and published the first X-ray diffraction pattern of DNA
- 1952 Roslaind Franklin shown better X-ray diffraction pattern of DNA



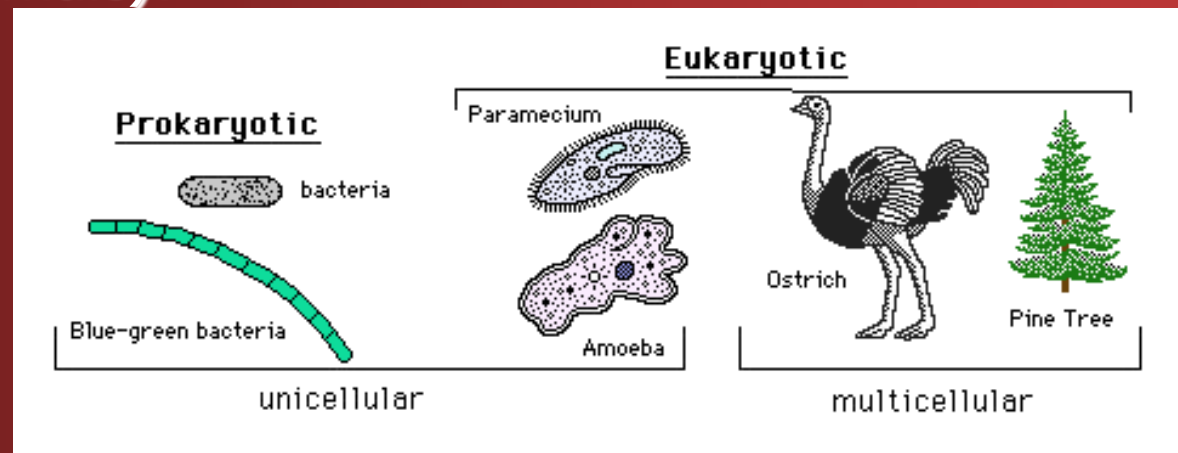
History of nucleic acids

- In 1953 Watson
- and Crick determined the structure of DNA



3- Cell Types

- All living organisms contain one or many cells.
- The organisms with only one cell in their body are called unicellular organisms (bacteria, blue green algae, some algae, Protozoa, etc.).
- The organisms having many cells in their body are called multi-cellular organisms (fungi, most plants and animals).

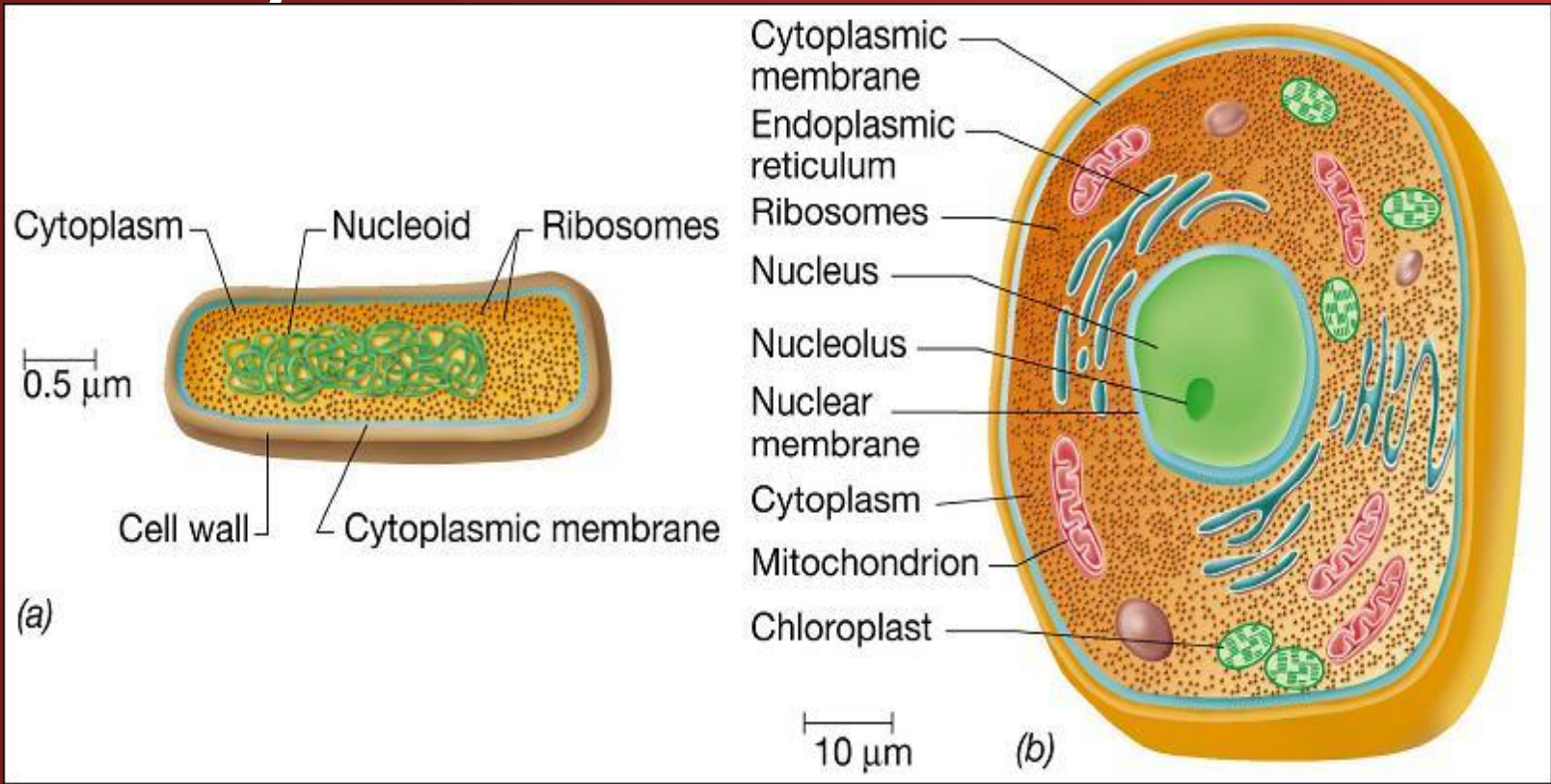


3- Cell Types

Any living organism may contain only one type of cell either

a- **Prokaryotic cells**

b- **Eukaryotic cells**

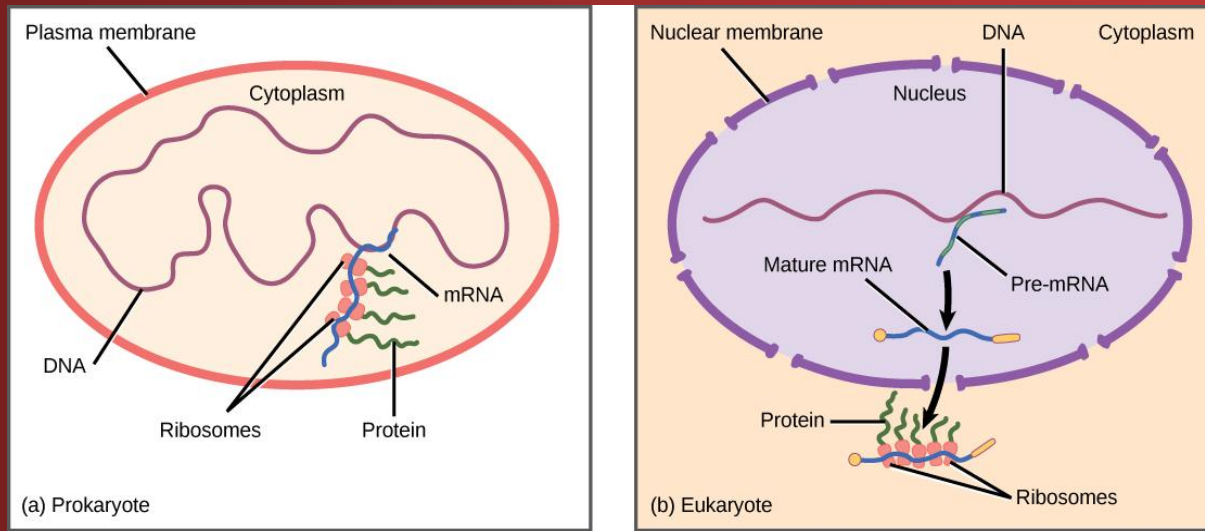


3- Pro. & eukaryotic cells Differences

Features	Prokaryotes	Eukaryotes
Nucleus	Absent	Present
Number of chromosomes	one	More than one
Shape of chromosomes	circular	Linear
Cell size	1 – 10um	10 – 100um
Mitochondria	Absent	Present
Chloroplast	Absent (chlorophyll found in the the cytoplasm)	Present
Golgi Apparatus	Absent	Present
Cell wall	Present	Present (in plants and fungi)
Cell type	Unicellular	Multicellular
Lysosomes	Absent	Present
Ribosomes	Small (70S)	Large (80S)
Cell Division	Binary fission	Mitosis and Meiosis
Examples	Bacteria	Animals and plants

4- Occurrences of nucleic acids

- Scientists succeeded in separating nucleic acids from the nuclei of cells.
- Nucleic acids are found in all living cells loose or united with the protein.
- Although first discovered within the nucleus of eukaryotic cells nucleic acids are now known to be found in all life forms including bacteria and viruses.



4- Nomenclature of nucleic acids

- The term nucleic acid is the overall name for DNA and RNA, members of a family of biopolymers, and is synonymous with polynucleotide.
- Nucleic acids were named for their initial discovery within the nucleus, and for the presence of phosphate groups (related to phosphoric acid).

Types of Nucleic acids

- The chemical studies on nucleic acids was done in the first place from two sources:
- on yeast, and found to contain the sugar ribose therefore named Ribonucleic acid (RNA)
- and the second on thymus gland of beef and found to contain the sugar deoxy-ribose, so named deoxyribonucleic acid (DNA).
- Kinds of Nucleic Acid: There are two general kinds of nucleic acids:
- Ribonucleic acid (RNA)
- Deoxyribonucleic acid (DNA)